Use-Case Flow Specifications (Multi-Use Case Package)

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07 — Campus & Academic Operations

UC-01: Log Maintenance Ticket (FR-001)

Primary Actor: Hostel Resident	**Supporting Actors:** Maintenance Warden
Preconditions: Resident logged into app with verified room tag.	**Postconditions (Success):** Ticket created, assigned unique ID, routed to Warden.

Main Success Scenario

1. Resident selects 'Log New Ticket' from the app home screen. System displays ticket form with fields for room number tag, category, and description.
2. Resident confirms or edits the pre-filled room tag (auto-detected from profile).
3. Resident selects an issue category (Plumbing, Electrical, Carpentry) and severity level (Low/Medium/High).
4. Resident optionally attaches a photo of the maintenance issue (Attach Photo Evidence extension).
5. Resident submits the ticket.
6. System validates that all required fields (room tag, category, severity) are present.
7. System generates a unique ticket ID, timestamps submission, and routes ticket to Maintenance Warden queue.
8. System displays confirmation screen with ticket ID and estimated response timeline. Use case ends.

Alternate Flow — 6a: Incomplete Submission

6a. Triggered at step 6 if a required field (e.g., room number tag) is missing or invalid.

6a1. System rejects submission and highlights missing field(s) in red.

6a2. System displays error prompt: 'Please complete all required fields before submitting.'

6a3. Resident inputs missing data and resubmits. Flow resumes at step 6 once validation passes.

UC-02: Assign Ticket & SLA (FR-002)

Primary Actor: Maintenance Warden	**Supporting Actors:** Maintenance Staff
Preconditions: Ticket exists in unassigned queue with 'Open' status.	**Postconditions (Success):** Ticket status updated to 'Assigned', staff notified.

Main Success Scenario

1. Warden logs into admin portal and navigates to 'Unassigned Ticket Queue'.
2. System displays list of newly submitted tickets sorted by severity (High priority on top).
3. Warden opens a ticket, reviews description, category, and room location.
4. Warden selects an available Maintenance Staff member from department roster.
5. System automatically populates recommended SLA target resolution timeframe based on issue severity.
6. Warden confirms assignment and submits.
7. System updates ticket status to 'Assigned', records warden assignment timestamp, and sends instant push/SMS notification to assigned staff member. Use case ends.

Alternate Flow — 4a: Staff Unavailable

4a. Triggered at step 4 if no maintenance staff are available for the selected category.

4a1. Warden flags ticket for External Vendor Dispatch.

4a2. System logs extended SLA timeline note and notifies Hostel Administration. Flow resumes at step 6.

UC-03: Update Ticket Status (FR-003)

Primary Actor: Maintenance Staff / Warden	**Supporting Actors:** Hostel Resident
Preconditions: Ticket is in 'Assigned' or 'In Progress' state.	**Postconditions (Success):** Status updated, timestamped, resident notified.

Main Success Scenario

1. Assigned Maintenance Staff or Warden accesses ticket details via mobile app or portal.
2. User selects new status update (e.g., transition from 'Open' → 'In Progress' upon arrival, or 'In Progress' → 'Resolved' after repair).
3. User optionally attaches work progress notes or photo of completed work.
4. User submits status update.
5. System validates transition logic, records server timestamp, and updates ticket status.
6. System sends automated status notification alert to reporting resident's dashboard and mobile app within 1 minute. Use case ends.

Alternate Flow — 2a: Spare Parts Delay

2a. Triggered at step 2 if repair requires specialized spare parts not currently in stock.

2a1. Staff sets status to 'On Hold — Awaiting Parts' and enters estimated delay.

2a2. System updates resident dashboard with delay explanation and revised resolution target. Use case ends.

UC-04: Record Closure Sign-off (FR-004 / Close Ticket)

Primary Actor: Hostel Resident	**Supporting Actors:** Maintenance Warden / Staff
Preconditions: Ticket status set to 'Resolved'.	**Postconditions (Success):** Resident sign-off recorded, status set to 'Closed'.

Main Success Scenario

1. System sends notification to resident once ticket status is marked 'Resolved'.
2. Resident inspects physical repair work completed in room.
3. Resident accesses ticket on app dashboard and selects 'Confirm Resolution & Sign-Off'.
4. Resident provides digital signature/confirmation and optional 1-5 star satisfaction rating.
5. System validates sign-off input, links resident ID + timestamp to ticket record, and updates status to 'Closed' (include relationship execution).
6. System archives ticket record into historical logs. Use case ends.

Alternate Flow — 3a: Resolution Rejected

3a. Triggered at step 3 if resident finds repair incomplete or unsatisfactory.

3a1. Resident selects 'Reject Resolution' and enters detailed feedback.

3a2. System reverts ticket status back to 'In Progress', notifies Warden queue, and increments escalation priority. Use case ends.

Requirements_Table

	A	B	C	D	E	F
1	Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
2	FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity (Plumbing/Electrical/Carpentry; Low/Medium/High).	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: Incomplete form (e.g., missing room tag) is accepted.	Core intake function; accurate tagging and severity are required to route and prioritize the ticket.
3	FR-002	Functional	The system shall allow the Maintenance Warden to assign a submitted ticket to a specific maintenance staff member and set a target resolution SLA based on issue severity.	High	Pass: Warden selects a staff member from a list, ticket status changes to "Assigned," and the assigned staff is notified within 1 minute. Fail: Ticket remains unassigned after the warden submits the assignment.	Ensures every logged issue is routed to a responsible person for timely resolution.
4	FR-003	Functional	The system shall allow the Maintenance Warden or assigned staff to update a ticket's status (Open → In Progress → Resolved) as work progresses.	High	Pass: Each status change is timestamped and visible on the resident's ticket view within 1 minute. Fail: A status change is not reflected to the resident.	Keeps residents informed and creates an auditable timeline needed for SLA tracking.
5	FR-004	Functional	The system shall require the reporting resident to confirm and record a closure sign-off before a ticket can be marked "Closed."	Medium	Pass: A ticket cannot transition to Closed without a stored sign-off (resident ID + timestamp). Fail: A ticket is marked Closed with no sign-off record.	Verifies the issue is actually resolved from the resident's perspective and creates accountability for maintenance staff.
6	FR-005	Functional	The system shall allow residents to view the real-time status and history of all maintenance tickets they have submitted.	Medium	Pass: The resident's dashboard lists every submitted ticket with current status and last-updated timestamp. Fail: A submitted ticket is missing from the resident's list.	Transparency reduces duplicate ticket submissions and manual status-check requests to the hostel office.
7	NFR-001	Nonfunctional — Performance & Security	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load, and escalation alerts fire within the 24-hour window.	Prevents high-severity issues (e.g., electrical hazards) from being silently ignored past the SLA.
8	NFR-002	Nonfunctional — Usability	The system shall allow a resident to complete and submit a new ticket, including a photo attachment, within 90 seconds on a standard smartphone browser.	Medium	Pass: Usability testing with 10 residents shows average submission time ≤ 90s. Fail: average exceeds 90s.	Low reporting friction increases adoption and reduces issues going unreported.

UseCase Diagram

